

Effect of VarroxCare Oxalic Acid Strip Placement on Varroa Mite Levels in Honey Bee Colonies

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Abstract

This study evaluated whether the placement of VarroxCare oxalic acid strips within honey bee colonies affected Varroa mite infestation levels during a 42-day summer treatment period. Twenty-four colonies located at two apiaries were divided equally among three groups: VarroxCare strips placed between brood frames according to labeled placement, VarroxCare strips placed horizontally on top of the brood frames, and untreated controls. Four VarroxCare strips were used per brood box in treated colonies from July 4 through August 15, 2026. Varroa infestation was measured before and after treatment using alcohol washes of approximately 300 adult worker bees collected from brood frames. Average infestation in the label-placement group changed from 0.42 to 0.38 mites per 100 bees, while the top-placement group increased from 0.25 to 0.63 mites per 100 bees. Untreated controls increased from 0.29 to 1.67 mites per 100 bees. No treated colony finished at or above 2 mites per 100 bees, compared with five of eight untreated control colonies. Although label placement produced a lower average post-treatment infestation than top placement, the difference between the two placement methods was not statistically significant. These results suggest that VarroxCare treatment limited Varroa population growth compared with untreated colonies during the study period, while this experiment did not provide sufficient evidence to conclude that one strip placement was more effective than the other.

Introduction

Varroa destructor continues to be one of the most serious threats facing managed honey bee (*Apis mellifera*) colonies. The mite reproduces within capped brood cells and parasitizes developing and adult honey bees. Varroa infestations can weaken individual bees and colonies and contribute to colony losses, making regular monitoring and control an important part of colony management (Rosenkranz, Aumeier, and Ziegelmann 2010).

Oxalic acid is one of several compounds used by beekeepers to manage *Varroa destructor*. One challenge with some oxalic acid applications is the presence of capped brood. Varroa mites reproduce within capped brood cells, where they are protected from oxalic acid treatments that primarily affect mites exposed on adult bees. Gregorc and Planinc (2002) found that oxalic acid treatments were considerably more effective during broodless periods than when capped brood was present. Other research has also demonstrated limitations in the effectiveness of conventional oxalic acid applications when brood is present (Jack, van Santen, and Ellis 2020).

Extended-release oxalic acid formulations offer a different approach by keeping oxalic acid-containing material within the colony for a longer period. Rather than relying on a single short exposure, the treatment remains available over an extended period while brood continues to emerge. Maggi et al. (2016) evaluated an extended-release oxalic acid formulation using four cellulose strips per colony in colonies containing brood. Their study followed colonies for 42 days and demonstrated that prolonged delivery of oxalic acid could provide effective Varroa control under their experimental conditions.

However, extended-release oxalic acid treatments have not produced consistent results in every study. Bartlett et al. (2023) found no evidence that an oxalic acid-glycerol formulation

delivered using paper towels reduced Varroa parasitism under their experimental conditions in the southeastern United States. More recently, Thurston et al. (2025) evaluated oxalic acid-glycerin strips during summer conditions in a northern climate and reported moderate efficacy, suggesting that this type of treatment may have value as part of an integrated Varroa management program.

Together, these findings indicate that the effectiveness of extended-release oxalic acid treatments can vary among studies and treatment conditions. Treatment formulation, environmental conditions, brood conditions, treatment duration, and delivery method are therefore important considerations when evaluating treatment performance.

VarroXSan is an extended-release oxalic acid treatment in which strips remain within the colony for an extended treatment period. The labeled placement positions the strips between brood frames. Because strip-based treatments rely on prolonged exposure within the colony, strip placement may influence how bees encounter the treatment material. Whether this difference in placement affects Varroa control was the focus of the present study.

The purpose of this study was to determine whether VarroXSan strip placement affected changes in Varroa mite infestation levels during a 42-day treatment period. Colonies receiving VarroXSan according to the labeled placement between brood frames were compared with colonies receiving the same number of strips placed horizontally on top of the brood frames and with untreated control colonies. Varroa infestation levels were measured using alcohol washes before and after the treatment period. The study was conducted at two separate apiaries to determine whether similar treatment patterns could be observed at different locations.

Research Question

Does placing VarroSan oxalic acid strips horizontally on top of brood frames, rather than between brood frames according to the labeled placement, affect Varroa mite infestation levels following a 42-day treatment period?

Materials and Methods

Experimental Design

Twenty-four honey bee colonies were included in the study at two separate apiaries, with 12 colonies located at each apiary. At each location, colonies were divided into three groups of four colonies each: a label-placement treatment group, a top-placement treatment group, and an untreated control group. Across both apiaries, this resulted in eight colonies per treatment group and 24 colonies overall.

Colonies in the label-placement group received four VarroSan oxalic acid strips per brood box positioned between the brood frames according to the labeled placement. Colonies in the top-placement group also received four VarroSan strips per brood box, but the strips were positioned horizontally on top of the brood frames. Colonies assigned to the control group received no VarroSan treatment.

Pretreatment Varroa mite levels were measured on July 3, 2026. VarroSan strips were installed in the treatment colonies on July 4, 2026, and remained in place through August 15, 2026, for a total treatment period of 42 days.

Varroa Sampling

Varroa mite infestation was measured using alcohol washes of approximately 300 adult worker bees, or about one-half cup of bees, collected from brood frames. The Honey Bee Health

Coalition recommends collecting approximately 300 adult bees from brood-nest combs for standardized Varroa sampling and identifies washing as a reliable method for estimating mite infestation (Honey Bee Health Coalition 2026).

A pretreatment alcohol wash was conducted for each of the 24 colonies before VarroXSan treatment began. Following the 42-day treatment period, a second alcohol wash was conducted on August 15 using the same sampling procedure. Only pretreatment and post-treatment alcohol-wash results were included in the analysis.

Mite counts were standardized as mites per 100 adult bees. Because approximately 300 bees were sampled from each colony, the number of mites recovered was divided by 300 and multiplied by 100.

Data Analysis

Changes in Varroa infestation levels from pretreatment to post-treatment were compared among the eight colonies receiving labeled VarroXSan placement, eight colonies receiving top-of-frame placement, and eight untreated control colonies. Results from the two apiaries were also examined separately.

Several colonies had zero or very low pretreatment mite counts. A zero count was not interpreted as evidence that a colony was completely free of Varroa mites because mites could have been present on unsampled adult bees or within capped brood (Honey Bee Health Coalition 2026).

Because a percentage reduction cannot be meaningfully calculated when the pretreatment mite count is zero, treatment performance was not evaluated solely by percentage reduction. Instead, pretreatment infestation, post-treatment infestation, absolute change in mites per 100

bees, final infestation relative to Varroa management thresholds, and differences among experimental groups were considered.

Post-treatment infestation levels were evaluated in relation to Varroa management guidance published by the Honey Bee Health Coalition (2026). Current guidance recommends management intervention when infestation levels exceed approximately one to two mites per 100 adult bees, while recognizing that appropriate thresholds may vary by region and colony conditions.

For colonies beginning with zero or very low detected mite levels, remaining below the management threshold at the end of the study was interpreted as maintaining a low infestation level rather than demonstrating a percentage reduction.

Because the treatment groups were relatively small and mite-count changes were not assumed to be normally distributed, changes among the three groups were compared using a Kruskal-Wallis test. Two-sided Mann-Whitney U tests were then used for pairwise comparisons. Statistical results were interpreted together with the descriptive results because of the limited sample size.

Results

All 24 colonies had pretreatment and post-treatment mite-wash results and were included in the final analysis.

Overall Treatment Results

The label-placement group began with an average of 1.25 mites per approximately 300-bee sample, equivalent to 0.42 mites per 100 bees. At the end of treatment, this group averaged

1.13 mites per sample, or 0.38 mites per 100 bees. The average change was a decrease of approximately 0.04 mites per 100 bees.

The top-placement group began with an average of 0.75 mites per sample, equivalent to 0.25 mites per 100 bees. At the end of the study, this group averaged 1.88 mites per sample, or 0.63 mites per 100 bees. This represented an average increase of approximately 0.38 mites per 100 bees.

Untreated control colonies began with an average of 0.88 mites per sample, equivalent to 0.29 mites per 100 bees. By August 15, controls averaged 5.00 mites per sample, or 1.67 mites per 100 bees. The control group therefore increased by approximately 1.38 mites per 100 bees during the study.

Table 1. Group-level Varroa infestation results

Treatment Group	Hives	Pre mites/100 bees	Post mites/100 bees	Average Change
Label placement	8	0.42	0.38	-0.04
Top placement	8	0.25	0.63	+0.38
Untreated control	8	0.29	1.67	+1.38

Treatment Thresholds

At the end of the study, none of the eight label-placement colonies and none of the eight top-placement colonies had a mite level of 2 mites per 100 bees or greater. In contrast, five of the eight untreated control colonies finished at or above 2 mites per 100 bees.

Using the more conservative level of 1 mite per 100 bees, two of eight label-placement colonies reached that level, three of eight top-placement colonies reached that level, and six of eight untreated control colonies reached or exceeded it. These results show a clear separation between the treated and untreated colonies by the end of the 42-day period.

Apiary Comparison

The same general pattern was observed at both apiaries. At Patrick's apiary, label-placement colonies averaged 0.42 mites per 100 bees before treatment and 0.25 after treatment. Top-placement colonies increased from 0.08 to 0.50 mites per 100 bees, while untreated controls increased from 0.42 to 2.00 mites per 100 bees.

At Doug's apiary, label-placement colonies averaged 0.42 mites per 100 bees before treatment and 0.50 after treatment. Top-placement colonies increased from 0.42 to 0.75 mites per 100 bees, while controls increased from 0.17 to 1.33 mites per 100 bees. Although the exact mite levels differed between apiaries, untreated colonies experienced the largest average increases at both locations.

Statistical Comparison

Changes in mite levels differed among the three experimental groups in the overall nonparametric comparison (Kruskal-Wallis test, $p = 0.025$). A pairwise comparison between the label-placement and top-placement groups did not show a statistically significant difference (Mann-Whitney U test, $p = 0.183$). Therefore, although the label-placement group had a lower average post-treatment mite level and smaller change than the top-placement group, the experiment did not provide sufficient evidence to conclude that label placement was statistically more effective than top placement.

The difference in mite-count change between the label-placement and untreated control groups was statistically significant (Mann-Whitney U test, $p = 0.017$). The comparison between top placement and untreated controls showed the same numerical direction but did not reach statistical significance ($p = 0.086$). Because only eight colonies were included in each group,

these statistical comparisons should be interpreted cautiously and together with the observed mite levels.

Discussion

The results of this study indicate that colonies treated with VarroSan experienced considerably less Varroa population growth during the 42-day treatment period than untreated control colonies. The untreated controls increased from an average of 0.29 to 1.67 mites per 100 bees, while label-placement colonies remained essentially stable at approximately 0.4 mites per 100 bees. Top-placement colonies increased from 0.25 to 0.63 mites per 100 bees but remained substantially below the final average observed among untreated controls.

These results are consistent with the general concept behind extended-release oxalic acid treatments. Maggi et al. (2016) demonstrated effective Varroa control using an extended-release oxalic acid formulation during a 42-day treatment period in colonies containing brood. The present study did not attempt to duplicate that formulation or calculate treatment efficacy using the same methodology, but the limited increase in mites among VarroSan-treated colonies compared with untreated controls supports the value of prolonged oxalic acid exposure during periods when brood is present.

At the same time, previous research has demonstrated that extended-release oxalic acid treatments do not perform consistently under every set of conditions. Bartlett et al. (2023) found no evidence that their glycerol-oxalic acid paper-towel treatment controlled Varroa under southeastern United States conditions, while Thurston et al. (2025) reported moderate efficacy from an oxalic acid-glycerin formulation during summer in a northern climate. These differences reinforce the importance of treatment formulation, delivery method, climate, brood conditions, and colony conditions when evaluating extended-release oxalic acid treatments.

The primary question of this study concerned strip placement. Label-placement colonies finished with an average infestation of 0.38 mites per 100 bees compared with 0.63 mites per 100 bees among top-placement colonies. Label-placement colonies remained essentially unchanged from their pretreatment level, while top-placement colonies experienced a modest increase.

This numerical difference suggests that placement between brood frames may provide an advantage, but the difference between the two VarroSan placement groups was not statistically significant. Therefore, the results do not demonstrate that one placement method was superior. A larger study with more colonies and higher initial mite levels would be needed to determine whether the numerical difference observed here represents a true placement effect.

The untreated controls provide important context for interpreting the low initial mite levels. Several colonies began with zero detected mites, making individual percentage-reduction calculations inappropriate. However, mite populations in untreated colonies increased considerably during the same 42-day period. Five of eight controls finished at or above 2 mites per 100 bees, whereas none of the 16 VarroSan-treated colonies reached that level.

This comparison is particularly useful because it demonstrates that low starting mite counts did not necessarily remain low without treatment. Colonies beginning with low or zero detected mites could still experience substantial Varroa population growth over the course of the study.

The two-apiary design also strengthens the practical interpretation of the experiment. Although mite levels differed between locations, the untreated controls showed the greatest average increase at both apiaries. This suggests that the overall treatment pattern was not limited to a single apiary.

There were several limitations to the study. Each treatment group contained only eight colonies, limiting statistical power. Initial mite infestation was also low in many colonies, which reduced the ability to measure large percentage reductions and created a floor effect for colonies beginning with zero detected mites. Alcohol washes represent samples of approximately 300 adult bees rather than a complete measurement of every mite within a colony, and mites present within capped brood are not included in the adult-bee wash.

The experiment also evaluated only one 42-day summer treatment period. Results could differ at another time of year, under different brood conditions, in a different climate, or when colonies begin with higher mite infestation levels. These limitations should be considered before applying the results broadly.

Conclusion

During this 42-day field study, Varroa mite populations increased substantially more in untreated control colonies than in colonies treated with VarroXSan. Label-placement colonies remained nearly unchanged, decreasing from an average of 0.42 to 0.38 mites per 100 bees, while top-placement colonies increased from 0.25 to 0.63 mites per 100 bees. Untreated controls increased from 0.29 to 1.67 mites per 100 bees.

None of the 16 VarroXSan-treated colonies finished at or above 2 mites per 100 bees, compared with five of eight untreated control colonies. This pattern was observed across both apiaries and indicates that VarroXSan treatment was associated with suppression of Varroa population growth during the study period.

Although the labeled between-frame placement produced numerically better results than horizontal top-of-frame placement, the difference between the two placement groups was not statistically significant. Therefore, this experiment does not provide sufficient evidence to

recommend changing the labeled placement or to conclude that one placement method is superior.

For beekeepers, the most important finding from this study is that both VarroSan treatment groups maintained substantially lower mite infestation levels than untreated controls during the 42-day summer period. Additional research using more colonies and colonies with higher initial mite levels would help determine whether strip placement has a meaningful effect on VarroSan performance.

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AI Disclosure

Artificial intelligence (ChatGPT) was used to assist with literature research, organization and editing of the technical report, verification of references and supporting sources, and

statistical analysis of the experimental data. The experimental design, field procedures, colony observations, and experimental data were provided and collected by the authors. The authors reviewed the analysis, interpretation, and conclusions presented in the final report.

Appendix A: Experimental Data

This appendix presents the colony assignments and raw pretreatment and post-treatment alcohol-wash counts used in the analysis. Each alcohol wash sampled approximately 300 adult bees. Standardized infestation values are expressed as mites per 100 bees. Percentage reduction is listed as N/A when the pretreatment count was zero because a percentage reduction cannot be calculated from a zero starting value.

Table A1. Colony-level experimental data

Hive	Apiary	Group	Placement	Pre mites 7/3	Post mites 8/15	Pre mites/ 100 bees	Post mites/ 100 bees	Change (post-pre)	Reduction %
P01	Patrick	Label	Between brood frames	2	0	0.67	0.00	-0.67	100%
P02	Patrick	Label	Between brood frames	0	2	0.00	0.67	+0.67	N/A
P03	Patrick	Label	Between brood frames	1	0	0.33	0.00	-0.33	100%
P04	Patrick	Label	Between brood frames	2	1	0.67	0.33	-0.33	50%
P05	Patrick	Top	On top of brood frames	0	1	0.00	0.33	+0.33	N/A
P06	Patrick	Top	On top of brood frames	0	0	0.00	0.00	+0.00	N/A
P07	Patrick	Top	On top of brood frames	0	2	0.00	0.67	+0.67	N/A
P08	Patrick	Top	On top of brood frames	1	3	0.33	1.00	+0.67	-200%
P09	Patrick	Control	No strips	1	3	0.33	1.00	+0.67	-200%
P10	Patrick	Control	No strips	3	6	1.00	2.00	+1.00	-100%
P11	Patrick	Control	No strips	0	8	0.00	2.67	+2.67	N/A
P12	Patrick	Control	No strips	1	7	0.33	2.33	+2.00	-600%
D01	Doug	Label	Between brood frames	0	3	0.00	1.00	+1.00	N/A
D02	Doug	Label	Between brood frames	0	0	0.00	0.00	+0.00	N/A
D03	Doug	Label	Between brood frames	2	3	0.67	1.00	+0.33	-50%
D04	Doug	Label	Between brood frames	3	0	1.00	0.00	-1.00	100%
D05	Doug	Top	On top of brood frames	0	1	0.00	0.33	+0.33	N/A
D06	Doug	Top	On top of brood frames	0	3	0.00	1.00	+1.00	N/A
D07	Doug	Top	On top of	2	2	0.67	0.67	+0.00	0%

Hive	Apiary	Group	Placement	Pre mites 7/3	Post mites 8/15	Pre mites/ 100 bees	Post mites/ 100 bees	Change (post-pre)	Reduction %
			brood frames						
D08	Doug	Top	On top of brood frames	3	3	1.00	1.00	+0.00	0%
D09	Doug	Control	No strips	1	1	0.33	0.33	+0.00	0%
D10	Doug	Control	No strips	0	8	0.00	2.67	+2.67	N/A
D11	Doug	Control	No strips	1	1	0.33	0.33	+0.00	0%
D12	Doug	Control	No strips	0	6	0.00	2.00	+2.00	N/A

Table A2. Group summary

Treatment group	n	Avg. pre mites/100	Avg. post mites/100	Average change
Label placement	8	0.42	0.38	-0.04
Top placement	8	0.25	0.63	+0.38
Untreated control	8	0.29	1.67	+1.38